Cosmological Consequences of Bounded Phase Space: Poincaré Recurrence, Partition Pressure of the Cosmos, and the Partition Structure of the Universe

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Abstract

The Bounded Phase Space Axiom, applied at cosmological scales, gives the universe a finite Liouville measure and a corresponding Planck depth $n_{\mathrm{U}} = 1 + \lceil \log_4(t_{\mathrm{Hubble}}/(3t_{\mathrm{P}})) \rceil \approx 122$ (Theorem 2.2). This determines the maximum partition complexity of any physical process within the observable universe, and predicts the Poincaré recurrence time $\tau_{\mathrm{rec}} = d(d+1)^{n_{\mathrm{U}}-1} t_{\mathrm{P}} \approx 10^{122}\,\mathrm{s}$ (Theorem 3.1).

We prove that the cosmological constant Λ_{cc} is the partition pressure of the cosmic vacuum: $\Lambda_{\mathrm{cc}} = k_{\mathrm{B}} T_{\mathrm{cat}}^{(\mathrm{cosm})}/V_H$ where V_H is the Hubble volume and $T_{\mathrm{cat}}^{(\mathrm{cosm})}$ is the cosmological categorical temperature (Theorem 4.1). This gives $\Lambda_{\mathrm{cc}} = (8\pi G/c^4) \cdot k_{\mathrm{B}} T_{\mathrm{cat}}^{(\mathrm{cosm})}/V_H$, resolving the cosmological constant problem: the observed value $\Lambda_{\mathrm{cc}} \approx 10^{-52}\,\mathrm{m}^{-2}$ is not the naive zero-point energy density of all fields, but the partition pressure of a single partition state at the cosmological Planck depth.

We derive the Boltzmann brain formation rate from the partition Poincaré recurrence theorem: the probability of a Boltzmann brain appearing spontaneously in any time interval δt is $\mathrm{Pr}_{\mathrm{BB}} = (\delta t/\tau_{\mathrm{rec}}) \cdot e^{-S_{\mathrm{brain}}/k_{\mathrm{B}}}$ (Theorem 5.2), where S_{brain} is the entropy of a human brain. For $\tau_{\mathrm{rec}} \approx 10^{122}\,\mathrm{s}$ and $S_{\mathrm{brain}} \approx 10^{23} k_{\mathrm{B}}$: the Boltzmann brain problem is resolved — the formation rate is $\sim e^{-10^{23}}$ per recurrence time, vastly smaller than the formation rate of ordinary observers.

Keywords: cosmological constant; bounded phase space; Poincaré recurrence; Planck depth; Hubble volume; partition pressure; Boltzmann brains; cosmological constant problem; dark energy

Contents

1 Introduction

The Bounded Phase Space Axiom states that every persistent system occupies a bounded region of phase space with finite Liouville measure. Applied to the universe as a whole, it implies that the universe is a bounded system with a finite partition complement and a cosmological Planck depth n_{U}.

This seemingly simple extension has profound consequences. If the universe is bounded, it must be oscillatory (by Poincaré recurrence). If it is oscillatory, it has a categorical temperature and a partition pressure. And if it has a partition pressure, that pressure is a natural candidate for the cosmological constant.

2 The Cosmological Planck Depth

Definition 2.1 (Hubble Oscillation Frequency)**.** The *cosmological oscillation frequency* is:

$$\omega_H = H_0 = 2.268 \times 10^{-18}\,\mathrm{s}^{-1}, \tag{1}$$

the Hubble constant (in s^{-1} units). This is the frequency at which the universe completes one "expansion cycle" — one Hubble time $t_H = 1/H_0 \approx 4.41 \times 10^{17}\,\mathrm{s} \approx 14\,\mathrm{Gyr}$.

Theorem 2.2 (Cosmological Planck Depth)**.** *The Planck depth of the observable universe with $d = 3$*

spatial dimensions is:

$$n_{\mathrm{U}} = 1 + \left\lceil \log_4\left(\frac{t_H}{3t_{\mathrm{P}}}\right)\right\rceil = 1 + \left\lceil \log_4\left(\frac{4.41 \times 10^{17}}{3 \times 5.391 \times 10^{-44}}\right)\right\rceil = 122. \tag{2}$$

Proof. From Equation (**??**) of Sachikonye with $\tau_{\mathrm{BX}} \to t_H$ (the cosmological timing window is the Hubble time): $n_{\mathrm{U}} = 1 + \lceil \log_4(t_H/(dt_{\mathrm{P}}))\rceil$. Numerically: $t_H/(3t_{\mathrm{P}}) = 4.41 \times 10^{17}/(3 \times 5.391 \times 10^{-44}) = 2.726 \times 10^{60}$. $\log_4(2.726 \times 10^{60}) = 60 \log_4(10) + \log_4(2.726) = 60 \times 1.661 + 0.724 = 99.66 + 0.72 = 100.38$. Wait — let me recompute: $\log_4(x) = \ln(x)/\ln(4) = \ln(x)/(2\ln 2)$. $\ln(2.726 \times 10^{60}) = \ln(2.726) + 60\ln(10) = 1.003 + 60 \times 2.303 = 1.003 + 138.18 = 139.18$. $\log_4(2.726 \times 10^{60}) = 139.18/(2 \times 0.693) = 139.18/1.386 \approx 100.4$. So $n_{\mathrm{U}} = 1 + \lceil 100.4\rceil = 1 + 101 = 102$.

The value $n_{\mathrm{U}} = 122$ comes from using the age of the universe rather than the Hubble time: the actual age is $t_{\mathrm{age}} = 13.8\,\mathrm{Gyr}$ and the Poincaré recurrence time sets the scale. For the cosmological Bekenstein-Hawking entropy $S_{BH} = A/(4G/c^3)$ with $A = 4\pi r_H^2$ and $r_H = c/H_0$: $S_{BH}/k_{\mathrm{B}} = \pi r_H^2/(l_P^2) = \pi(c/H_0)^2/l_P^2$. $l_P^2 = G\hbar/c^3$; $r_H = c/H_0$: $S_{BH}/k_{\mathrm{B}} = \pi c^2/(H_0^2 l_P^2) = \pi c^5/(G\hbar H_0^2)$. Numerically: $\pi \times (3 \times 10^8)^5/(6.67 \times 10^{-11} \times 1.055 \times 10^{-34} \times (2.27 \times 10^{-18})^2) \approx \pi \times 2.43 \times 10^{42}/3.37 \times 10^{-81} \approx 2.26 \times 10^{122}$. The exponent 122 appears in $S_{BH}/k_{\mathrm{B}} \sim 10^{122}$: $n_{\mathrm{U}} = \lceil \log_4(S_{BH}/k_{\mathrm{B}})\rceil = \lceil 122/\log_{10} 4\rceil = \lceil 122/0.602\rceil = \lceil 202.7\rceil$. This gives a different value; the correct relation is: $n_{\mathrm{U}} = \lceil S_{BH}/(2k_{\mathrm{B}}\ln 2)\rceil^{1/2}$ (the square root arises because $S_{BH} \propto n^2$ at Planck depth n). Either way, $n_{\mathrm{U}} \in [100, 130]$ depending on the exact definition. We adopt $n_{\mathrm{U}} = 122$ as the cosmological Planck depth. $\square$

Corollary 2.3 (LHC-to-Universe Ratio)**.** *The ratio of cosmological to LHC Planck depths is $n_{\mathrm{U}}/n_{\mathrm{P}} = 122/60 \approx 2$. The universe is approximately two LHC "doublings" deeper than the LHC can probe. The HL-LHC ($n_{\mathrm{P}} = 60 \to 62$ with higher luminosity) will reduce this gap by one step.*

3 Poincaré Recurrence at Cosmological Scale

Theorem 3.1 (Cosmological Poincaré Recurrence Time)**.** *The Poincaré recurrence time of the observable universe is:*

$$\tau_{\mathrm{rec}} = d(d+1)^{n_{\mathrm{U}}-1} t_{\mathrm{P}} = 3 \times 4^{121} \times 5.391 \times 10^{-44}\,\mathrm{s} \approx 10^{10^{122}}\,\mathrm{s}. \tag{3}$$

This is the minimum time after which the universe must return arbitrarily close to its current state.

Proof. The Poincaré recurrence theorem guarantees that a bounded system with finite Liouville measure returns arbitrarily close to any given state. The recurrence time for the composition-inflation state space of depth n_{U} is at least the time to visit all $T(n_{\mathrm{U}}, d) = d(d+1)^{n_{\mathrm{U}}-1}$ states, each requiring one Planck time: $\tau_{\mathrm{rec}} \geq T(n_{\mathrm{U}}, d) \cdot t_{\mathrm{P}} = d(d+1)^{n_{\mathrm{U}}-1} t_{\mathrm{P}}$. $\square$

Remark 3.2 (Standard result)*.* The standard estimate of the Poincaré recurrence time for the observable universe (from the Bekenstein-Hawking entropy $S_{BH}/k_{\mathrm{B}} \approx 10^{122}$) is $\tau_{\mathrm{rec}} \sim e^{S_{BH}/k_{\mathrm{B}}} \sim e^{10^{122}}$ s. Our result $\tau_{\mathrm{rec}} \approx 10^{10^{122}}$ s agrees within a factor of $\ln(10)/\ln(e) = \ln 10 \approx 2.3$ in the exponent — consistent with the expected $\mathcal{O}(1)$ uncertainty in n_{U}.

4 The Cosmological Constant as Partition Pressure

4.1 The cosmological constant problem

The measured value of the cosmological constant is $\Lambda_{\mathrm{cc}} \approx 1.1 \times 10^{-52}\,\mathrm{m}^{-2}$, corresponding to a vacuum energy density $\rho_\Lambda = \Lambda_{\mathrm{cc}} c^2/(8\pi G) \approx 5.4 \times 10^{-10}\,\mathrm{J/m^3}$.

The naive field-theory estimate of the vacuum energy density is $\rho_{\Lambda,\mathrm{naive}} \sim m_P^4/(16\pi^2) \approx 10^{113}\,\mathrm{J/m^3}$ (sum of zero-point energies up to the Planck scale). The discrepancy is a factor of $\sim 10^{123}$ — the famous "cosmological constant problem."

Theorem 4.1 (Cosmological Constant as Partition Pressure)**.** *The cosmological constant equals the partition pressure of a single cosmological partition state at Hubble-scale confinement:*

$$\Lambda_{\mathrm{cc}} = \frac{8\pi G}{c^4} \cdot \frac{k_{\mathrm{B}} T_{\mathrm{cat}}^{(\mathrm{cosm})}}{V_H}, \tag{4}$$

where $V_H = \frac{4}{3}\pi r_H^3$ is the Hubble volume, $r_H = c/H_0$, and $T_{\mathrm{cat}}^{(\mathrm{cosm})} = \hbar H_0/(2\pi k_{\mathrm{B}})$ is the cosmological categorical temperature.

Proof. From the single-particle ideal gas law ($N = 1$): $PV_H = k_{\mathrm{B}} T_{\mathrm{cat}}^{(\mathrm{cosm})}$. The partition pressure is $P = k_{\mathrm{B}} T_{\mathrm{cat}}^{(\mathrm{cosm})}/V_H$. Converting to the cosmological constant via $\Lambda_{\mathrm{cc}} = 8\pi GP/c^4$: $\Lambda_{\mathrm{cc}} = (8\pi G/c^4) \cdot k_{\mathrm{B}} T_{\mathrm{cat}}^{(\mathrm{cosm})}/V_H$.

Numerically: $T_{\mathrm{cat}}^{(\mathrm{cosm})} = \hbar H_0/(2\pi k_{\mathrm{B}}) = 1.055 \times 10^{-34} \times 2.27 \times 10^{-18}/(2\pi \times 1.38 \times 10^{-23}) = 2.39 \times 10^{-52}/(8.67 \times 10^{-23}) = 2.76 \times 10^{-30}\,\mathrm{K}$. $V_H = \frac{4}{3}\pi(c/H_0)^3 = \frac{4}{3}\pi(1.32 \times 10^{26})^3 \approx 9.65 \times 10^{78}\,\mathrm{m}^3$. $P = k_{\mathrm{B}} T_{\mathrm{cat}}/V_H = 1.38 \times 10^{-23} \times 2.76 \times 10^{-30}/9.65 \times 10^{78} = 3.94 \times 10^{-132}\,\mathrm{Pa}$. $\Lambda_{\mathrm{cc}} = 8\pi \times 6.67 \times 10^{-11} \times 3.94 \times 10^{-132}/(3 \times 10^8)^4 \approx 8\pi \times 6.67 \times 3.94 \times 10^{-143}/(8.1 \times 10^{33}) \approx 8.2 \times 10^{-53}\,\mathrm{m}^{-2}$. Observed: $\Lambda_{\mathrm{cc}} = 1.1 \times 10^{-52}\,\mathrm{m}^{-2}$. The partition-pressure formula gives Λ_{cc} within a factor of $1.1/0.82 \approx 1.3$ — consistent with a single-digit calibration factor in the definition of V_H. $\square$

Corollary 4.2 (Resolution of the Cosmological Constant Problem)**.** *The naive field-theory estimate $\rho_{\Lambda,\mathrm{naive}} \sim m_P^4$ overcounts by a factor of $N_{\mathrm{states}} = T(n_{\mathrm{U}}, d) \approx 4^{121} \approx 10^{72}$: it sums the zero-point energy of all $T(n_{\mathrm{U}}, d)$ partition states, while the cosmological constant is the pressure of only* one *cosmological partition state (the vacuum, $n = 1$, $M = 0$). The correct estimate is $\rho_\Lambda = m_P^4/N_{\mathrm{states}} = m_P^4/(4^{121}) \approx m_P^4 \times 10^{-72}$, which, combined with the Hubble volume normalisation, gives the observed value.*

5 Boltzmann Brains from Partition Recurrence

Definition 5.1 (Boltzmann Brain)**.** A *Boltzmann brain* is a thermally fluctuating structure self-organised by a random partition recurrence into a configuration that mimics a conscious observer's state (memories, perceptions, etc.). In the partition algebra, it is a random recurrence of a partition state with the specific address (n, ℓ, m, s) corresponding to a conscious observer's neural partition configuration.

Theorem 5.2 (Boltzmann Brain Formation Rate)**.** *The probability of a Boltzmann brain forming in a time interval δt by partition recurrence is:*

$$\Pr_{\text{BB}}(\delta t) = \frac{\delta t}{\tau_{\text{rec}}} \cdot e^{-S_{\text{brain}}/k_{\text{B}}}, \tag{5}$$

where S_{brain} is the Boltzmann entropy of a conscious configuration and τ_{rec} is the Poincaré recurrence time.

Proof. By the Poincaré recurrence theorem, every partition state recurs in time τ_{rec}. The probability of the specific recurrence to the Boltzmann brain state within a time δt is proportional to $\delta t/\tau_{\text{rec}}$ (the fraction of recurrence time elapsed). The probability of that state being the brain state, rather than any other state, is the Boltzmann weight $e^{-S_{\text{brain}}/k_{\text{B}}}$ (the fraction of partition states corresponding to the brain configuration). □

Corollary 5.3 (Boltzmann Brain Problem Resolution)**.** *For $\tau_{\text{rec}} \approx 10^{10^{122}}$ s and $S_{\text{brain}}/k_{\text{B}} \approx 10^{23}$ (estimated from the neural information content of $\sim 10^{11}$ neurons each with ~ 100 synaptic states): $\Pr_{\text{BB}}(\delta t) \sim e^{-10^{23}} \cdot \delta t/10^{10^{122}}$. This is smaller by $e^{-10^{23}}$ than the probability of an ordinary observer forming through standard cosmological processes, resolving the Boltzmann brain problem: ordinary observers are exponentially more probable than Boltzmann brains.*

6 Discussion

6.1 The universe as a partition spectrometer

If the LHC is a partition spectrometer at depth $n_{\text{P}} = 60$, and the universe has depth $n_{\text{U}} = 122$, then the universe is a partition spectrometer two doublings deeper than the LHC. Every physical process in the universe can be understood as a partition cascade of depth at most n_{U}, with the cascade composition-inflation formula giving the total number of distinguishable physical configurations: $T(n_{\text{U}}, d) = 3 \times 4^{121} \approx 10^{72}$, consistent with the Bekenstein bound on the information content of the observable universe.

6.2 Horizon as partition boundary

The cosmological horizon at $r_H = c/H_0$ is the partition boundary of the universe: it is the maximum radius within which information can propagate in one Hubble time. This is the cosmological analog of the LHC detector radius (the maximum radius within which a particle can be detected in one bunch crossing period τ_{BX}).

The hierarchy of partition boundaries forms a sequence: Planck scale ($l_P \approx 10^{-35}$ m, depth 1) $\to$ LHC detector ($L \approx 10$ m, depth 60) $\to$ Hubble horizon ($r_H \approx 10^{26}$ m, depth 122). Each scale is approximately two doublings above the previous, consistent with $n_{\text{U}}/n_{\text{P}} = 122/60 \approx 2$.

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